Lab 06-Regression and Classification

ITAI 1371

Team BetaThetas

Reflection Journal

Brandy Gonzales

Before this lab, I didn't quite get the concept of MSE or why it would be necessary in data analysis. What made it click was seeing the model guess $14.50 when the actual fare was $14.45. Accuracy would count that as wrong, even though it was very close. It's important to know not just whether your model guessed correctly, but how far off the mark it was.

One challenge I ran into was a NaN error. I ran the cell that changes Sex into numbers twice, so the second time it couldn't find "male" or "female" anymore and turned everything into blanks. I fixed it by restarting the session and running all the cells once. I learned that a cell that changes the data should only be run one time.

This lab helped me see the difference between regression and classification. Regression predicts a number, like Fare, and is measured with MSE. Classification predicts a category, like Survived, and is measured with accuracy. My logistic model was 74.83% accurate, and my fare guesses were typically off by about $58 (RMSE).

Hey everybody, Here is my reflection and contribution journals for Lab 06. If you all need the completed Jupiter notebook. I have that too. Just let me know.

Derrick K. Gibbs-McGlaston Lab 06 Reflection Journal

Working through this lab helped me understand why regression and classification are different, even though both are forms of supervised learning. The distinction became clearer when I used the same Titanic dataset to answer two different questions. Predicting a ticket fare means predicting a number, so linear regression fits that problem. Predicting whether a passenger survived means choosing between categories, so logistic regression fits that problem. I can now see that the question I want answered should guide the model I select. I also have a better understanding of why preparing data matters before training a model. The notebook removed records with missing ages, selected the columns needed for each task, and separated the data into training and testing groups. Both tasks used 571 records for training and 143 for testing. The models learned from one group of records and were checked against another. This made the workflow easier for me to follow because each step had a clear purpose. The results taught me not to confuse a working program with a perfect prediction. The linear regression model produced a mean squared error of 3364.92 and a root mean squared error of about $58.01. Seeing the error expressed in dollars helped me understand that a model can run successfully and still make predictions that are far from the actual fares. The coefficients also showed how the model used age and passenger class, but those numbers alone do not establish why a fare was higher or lower. The classification model predicted survival correctly for 107 of the 143 test records, or 74.83%. That result was easier to interpret than the fare error, but it also raised a question for me: what kinds of mistakes did the model make on the other 36 records? Accuracy gives an overall result, but it does not explain every incorrect prediction. This is something I would like to examine more closely in a future lab. The biggest lesson I took from this exercise is that machine learning is a process, not just a model. I have to understand the problem, choose useful data, prepare it, train a model, test it, and decide what the results actually mean. The tools used in earlier labs are useful because they make those steps possible, but the tools do not make the decisions for me. I need to understand what I am asking the model to predict and whether the result is useful.

Reflective Journal

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Course: ITAI 1371

In this lab, I learned the difference between regression and classification, two important types of supervised machine learning. Regression is used when the goal is to predict a continuous numerical value, while classification is used when the goal is to predict a

category or class. The notebook demonstrated both ideas using the Titanic dataset, which made the difference easier for me to understand.

For the regression part, I used Age and Pclass to predict Fare. I created and trained a Linear Regression model, then used the test data to make predictions. The model was evaluated using Mean Squared Error, and my result was **3364.92**. This helped me understand that regression models are evaluated by measuring how far the predicted numerical values are from the actual values. I also examined the model coefficients, which were about **-0.44 for Age** and **-38.30 for Pclass**. This showed me that the model can also help explain how different features influence the prediction.

For the classification part, I used Age, Pclass, and Sex to predict whether a passenger survived. I trained a Logistic Regression model and evaluated it using accuracy. The model achieved **74.83% accuracy**, which means it correctly predicted survival for about three out of every four passengers in the test data. This helped me understand why classification problems use a different evaluation method from regression problems.

One of the most important things I learned is that the type of target variable determines the type of machine learning problem. If the target is a number, such as fare, the problem can be treated as regression. If the target is a category, such as survived or not survived, the problem is classification. I also learned that the evaluation metric must match the type of problem. Mean Squared Error is useful for numerical predictions, while accuracy is useful when comparing predicted classes with actual classes.

Overall, this lab improved my understanding of supervised machine learning and helped me see how models are trained, tested, and evaluated. I now have a clearer understanding of how Linear Regression and Logistic Regression are used for different types of prediction problems, and why choosing the correct model and evaluation method is important.

Marlon Roberts- Im learning to be more confident putting in more valid data sets and knowing how to do the data preparations. Everything is slowly coming into process, the more knowledge I know, the more I will flow with data analysis.